EXPERIMENT 5

Lab Experiment: Design and Develop an Interactive User Interface using React

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Aim

To design and develop an **interactive Student Portfolio User Interface** using **React** by creating reusable components and rendering dynamic content.

Learning Objectives

After completing this experiment, students will be able to:

- Understand the basics of React
- Create React applications using Create React App
- Develop functional components
- Use JSX syntax
- Understand props and state
- Run and test a React application

Software / Tools Required

- Operating System: Windows / Linux
- Node.js (LTS version)
- npm (Node Package Manager)
- Visual Studio Code
- Web Browser (Chrome / Firefox)

Pre-requisites

- Basic knowledge of HTML, CSS, and JavaScript
- Node.js installed and verified using node -v

Theory

React is a JavaScript library developed by Facebook for building user interfaces, especially single-page applications. React allows developers to create reusable UI components and efficiently update and render components when data changes.

In this experiment, a Student Portfolio UI is developed using React components such as Header, About, Skills, Projects, and Footer.

Folder Structure

```
react-ui/
|
└── student-portfolio/
 ├── public/
└── src/
 ├── components/
| ├── Header.js
| ├── About.js
| ├── Skills.js
| ├── Projects.js
| └── Footer.js
 ├── App.js
└── index.js
```

Algorithm

1. Install Node.js and npm.
2. Create a new React application.
3. Create reusable components.
4. Import components into App.js.
5. Render components in the browser.
6. Verify UI output.

Step-by-Step Procedure

Step 1: Install Node.js

1. Download Node.js from official website.
2. Verify installation using commands:

```
node -v

npm -v
```

Step 2: Create React Application

1. Open terminal in VS Code.
2. Run the following command:

```
npx create-react-app student-portfolio
```

3. Navigate into project folder:

```
cd student-portfolio
```

Step 3: Start React Development Server

```
Run:  npm start
```

Verify default React page opens in browser.

Step 4: Create Components Folder

1. Inside src, create folder named components.
2. Create component files.

Step 5: Create Header Component (Header.js)

```
function Header() {

return (

<header>

<h1>Student Portfolio</h1>

<p>Computer Engineering Student</p>

</header>

);

}

export default Header;
```

Step 6: Create About Component (About.js)

```
function About() {
return (
<section>
<h2>About Me</h2>
<p>I am a third year Computer Engineering student interested in web development.</p>
</section>
);
}
export default About;
```

Step 7: Create Skills Component (Skills.js)

```
function Skills() {
return (
<section>
<h2>Skills</h2>
<ul>
<li>HTML</li>
<li>CSS</li>
<li>JavaScript</li>
<li>React</li>
</ul>
</section>
);
}
export default Skills;
```

Step 8: Create Projects Component (Projects.js)

```
function Projects() {
return (
```

```
<section>
<h2>Projects</h2>
<p>Student Portfolio Management System</p>
</section>
);
}
export default Projects;
```

Step 9: Create Footer Component (Footer.js)

```
function Footer() {
return (
<footer>
<p>© 2025 Student Portfolio</p>
</footer>
);
}
export default Footer;
```

Step 10: Update App.js

```
import Header from './components/Header';
import About from './components/About';
import Skills from './components/Skills';
import Projects from './components/Projects';
import Footer from './components/Footer';
function App() {
return (
<div>
<Header />
<About />
<Skills />
```

```
<Projects />
<Footer />
</div>
);
}
export default App;
```

Screenshots to Capture

- Node.js version output
- React project creation command
- Folder structure in VS Code
- React UI running in browser

Expected Output

An interactive Student Portfolio user interface rendered using React components.

Assignment / Submission

- Upload react-ui folder to GitHub repository
- Commit message: React portfolio UI

Viva / Review Questions

1. What is React?
2. What is JSX?
3. Difference between props and state?
4. What is a component?

Conclusion

In This Experiemnt an interactive Student Portfolio UI was successfully developed using React.